

Protecting Salem's Ecosystem: Predictive Modelling of Tree Canopies and Stream Temperatures

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Abstract

Rising stream temperatures are threatening Salem's ecosystem due to many environmental factors. The loss in tree canopy coverage and increasing temperatures risk pushing native species to cooler bodies of water and away from Salem's waterways. Here, we aimed to model the change in Salem's Mill Creek temperature from 2015 to 2025 using historical stream temperature, weather, and canopy coverage data using linear regression models. One model stood out among the rest, with an adjusted R-squared value of 0.9442. This model combines high-level interactions within temperature and time to predict stream temperatures. Within the model, with the available data, tree canopy does not have a significant impact on stream temperature. The main driver of the model is time. This means that, as it stands, more work should be done in protecting stream temperature against future rising temperatures, regardless of tree canopy coverage. A defining challenge in this project included a lack of reliable data; thus, it is also recommended that the City of Salem allocate more resources to collecting data that would shed more light on factors impacting stream temperature.

Introduction and Background

Bodies of water around the world are beginning to heat up at faster rates than air heat waves (Bush, 2025). These water heatwaves—defined by at least five consecutive days above the seasonal average—result in lower dissolved oxygen content, nutrient cycling, and pH buffering (PennState, 2026). All of this means that ecosystems and native species that have adapted to live in colder waters face serious threats.

One way to combat high stream temperatures is to support riparian zones, the dense plant areas along rivers (University of Georgia, 2026). These areas often sport trees and other foliage

that shade and cool rivers and streams, effectively lowering the water temperature. However, these areas are at risk for several reasons.

The first risk to riparian zones is human development built alongside waterways (University of Georgia, 2026). If projects such as Salem's proposed Riverfront Redevelopment Site are not done explicitly with the ecosystem in mind, there can be negative consequences due to deforestation and disruptions to the riparian zones.

Secondly, Emerald Ash Borers are a new concern in Salem. Emerald Ash Borers are pests that are native to Northern China, Russia, Japan, and Korea (USDA, 2023). These insects are highly corrosive to North American Ash Trees and are killing trees across the United States (Penn State Extension, 2025). These beetles were found in Oregon for the first time in 2022 and have been steadily increasing their lifespan as the climate changes (Boboc, 2026). This recent expansion caused Oregon to start thinking critically about readiness and response plans, including Salem's own plan (City of Salem, 2026).

With these complex system problems in mind, four research questions were developed:

1. Can historic stream temperature data be used to predict stream temperature with an R-squared value of at least 90%?
 - a. Will adding historic weather data for Salem, Oregon improve the R-squared value from the original model?
 - b. Will adding historic tree canopy data for Salem, Oregon improve the R-squared value from the original model?
2. Can a model be used to predict Salem's stream temperature 3 years into the future based on different levels of anticipated climate change impact on the weather data?

The vision for this project is to help government and private organizations prevent the loss or disruption of riparian zones in Salem. By developing an effective model that can offer insights into trends in stream temperature and its potential effects on Salem's waterways, this project will seek to define the severity of this issue and highlight targeted preventative measures that can be taken. This is a topic that the City of Salem, or private researchers, have not prioritized due to a lack of resources and the very recent appearance of the Emerald Ash Borers. This means that there is no previous research on this topic in Salem, Oregon.

This project is of particular interest for our group because our group members were born and raised in the Willamette Valley, surrounded by nature and a curiosity for stewardship. This unification brings a personal element to the heart of the project.

Impact and stakeholders

This project was outlined with the City of Salem's input and feedback. This topic was of interest to our group members but was also raised by the City of Salem due to their stake in this topic but lack of resources and man-power to perform the analysis themselves. The results of this project will thus be reported to and benefit the City of Salem. Using these results, a more informed plan can be created by the City of Salem to protect riparian zones in Salem, along with the indigenous species that live in them.

The Shared Theme domain for our project is the Pacific Northwest forest and environmental health. Our Peer Project Owners are looking into fire patterns in Oregon, and could provide different perspectives about topics such as tree weaknesses from drought or heat, as well as heat ecology damages in general that may compound our data. This additional perspective will allow this project to consider a more holistic view of rising stream temperatures, their causes, and impacts.

Data and Data Engineering

Data Collection and Cleaning

The first data source that we utilized for our analysis was collected by CastawayWU as part of the Mill Creek Research Project. This dataset contains daily average temperatures, dissolved oxygen readings, and turbidity for two sites along the Mill Creek. The two data collection sites were MIC3, in downtown Salem, and MIC12, located southwest of downtown Salem. This dataset was sent to us as a comma-delimited file (CSV) with 7543 rows and 5 columns, and had a file size of 426 KB. The raw dataset is available by request to Dr. Gore or CastawayWU.

Before this data could be used in prediction models, we performed data cleaning in an RMarkdown to get the most reliable results and add feature engineering. The first modification to the dataset was to remove the rows of the data prior to January 1st of 2015. This was necessary since there were only two rows of data from 2014, which wouldn't have supported our models. Then, the day of the year, year, and month were added as columns to the dataset to give the models more pieces of information to utilize.

The second piece of data that we brought into our project was climate data for Salem from NOAA. This dataset gave us daily information about the location of data collection, the average wind speed, precipitation, if it snowed, snow depth, highest temperature, lowest temperature, and average temperature. This dataset was downloaded as a CSV file with 48603 rows and 18 columns, and had a file size of 15.1 MB. To retrieve this data, we added filters for only Daily Summaries from a date range of 1900-01-01 to 2026-06-30, and only requested information collected by the McNary Field station.

To clean this dataset for analysis, the main task was to retain only the columns that were relevant to our research. These columns provide the information previously cited and have the column names: STATION, NAME, LATITUDE, LONGITUDE, DATE, AWND, PRCP, SNOW, SNWD, TMAX, TMIN, and TAVG. Then, rows with missing data in TMAX, TMIN, PRCP, and DATE were filtered out of the final dataset.

An important note about this dataset is that the null values don't mean that the values here should be zero. Zero is entered when there is a record for the different weather recording variables, not null. We are not entirely sure why these null values occur, but a majority seem to come from older dates than newer ones.

The third data source used in this project contained information on public storm creeks and rivers in Salem from the City of Salem and was hosted on ArcGIS. This dataset contained the locations of all bodies of water in Salem, including the two data collection sites of the Mill Creek from the stream temperature data. This data was downloaded as a shapefile with 2,672 rows and 24 columns, and had a file size of 619 KB. This data can be downloaded from ArcGIS.

This shapefile did not necessitate any cleaning for its purposes. This data will only be used to determine the tree density around the two data collection sites but will not be directly used in the prediction models. Additionally, there were no missing values or outliers, and no columns were removed.

The final data source used in our research was tree canopy information from the Tree Plotter Canopy website managed by the City of Salem. This data contained polygons for different levels of tree canopy area across the Salem area. This data was downloaded as a shapefile with 3,649 rows and 18 columns, and had a file size of 3,184 KB. The map is publicly

available; however, a City of Salem login is required to download the raw data. To retrieve this data, request it from the City of Salem.

Preparing this data for analysis was more lengthy than the other data sources. The process for generating the final tree canopy dataset uses two raw files: the cleaned creek locations shapefile (creek_locations.shp) and the raw tree canopy shapefile downloaded from the Tree Canopy website (raw_urban_tree_canopy.shp). Both of these shapefiles were loaded into ArcGIS Pro.

First, the Buffer tool in ArcGIS Pro was used to create a 100ft buffer on the streams to represent the riparian zone. This 100ft buffer represents the geographic areas we are going to focus on for tree canopy along streamsides. Next, this buffer layer was used in the Clip tool to clip the tree canopy shapefile, so that the tree canopy data includes only the 100ft streamside riparian zones. Next, the Selection tool was used to select only the stream represented by the MIC3 and MIC12 stream temperature measurement sites. The stream segments corresponding to each of these measurement sites are of similar length. From this layer, the attribute table was exported. Separate attribute tables for MIC3 and MIC12 were exported as well so that the two creek segments can be modeled separately. All of these files are attached in the repository for easy reference, as each step was conducted in ArcGIS Pro.

After the GIS processing was complete, the MIC3 and MIC12 tables were wrangled in R to form the final datasets for both creek segments. The tables were converted into long format, creating the variable UrbanTreeCanopy_pct as the percentage of urban tree canopy as observed that year. Unnecessary columns were removed, keeping 9 columns: Impervious, Water, Herbaceous, BareSoilDr, Unsuitable, PossiblePl, Shrub, Year, and UrbanTreeCanopy_pct. This was then used to create median aggregates for each year for all the variables for MIC3 and

MIC12 creek locations, creating two final modeling datasets: mic3_summary_by_year and mic12_summary_by_year. For other visualization purposes, the non-aggregated versions of the datasets were also saved: tree_canopy_mic3.csv and tree_canopy_mic12.csv.

Data Engineering

The data used in this project was contained in stagnant CSV files or Shapefiles which meant that no continuous refreshes or scraping was needed. This decision was made because the stream temperature and tree canopy coverage data were not sources that were frequently updated with new data. The weather data from NOAA was the only data source that was continuously updated but the new information would not have been pertinent to our research without updated stream temperature data on the same dates.

The stream temperature, weather, and tree canopy coverage data that was collected from our sources was moved into a PostgreSQL server hosted by Railway. This allowed each group member to access the data when needed. As the data was being moved into the database, a normalized relational database was created to ensure that all data sources were cleaned and could be joined successfully in the analysis steps.

The majority of the analysis for this project was completed in R. Thus, to access the data, commands such as /COPY were used to load the data into CSV files on personal devices. This meant that copies of the original data were stored in each member's file system to be easily accessible for analysis in R.

Ethics and Responsible Use

Concern about predictability of data.. Some randomness in weather + Removing outliers

The main ethical concern in this research is the accuracy of the model that was created. Despite using best practices and data sources being marketed as reliable, it is possible that the

model will generally overestimate or underestimate the change in stream temperature. There tends to be some natural randomness in the weather and stream temperature data, which leads to it being more challenging to predict. In the exploratory data analysis process, it was noted that there were multiple outliers in the stream temperature and weather datasets due to record-breaking temperatures, which might have disproportionately influenced the model towards higher temperatures. To reduce this skew, outliers were removed from the dataset; however, there may have been other similar oddities in the data that our team failed to identify and manage.

While there are ethical concerns regarding this research and its implications, there were no concerns regarding data accessibility and sharing personally identifiable data. Data collected and displayed by the City of Salem is available freely by request, and the CastawaysWU data is also accessible by request. None of our data sources contain information about individuals; thus, sharing this data would not lead to any breach of privacy or confidentiality.

For the most responsible use of this data, users should reference the predictions alongside domain expertise to identify areas to expend preventative measures. The models created in this project should not be used as the one source of truth, despite high accuracy measures.

Methods and Evaluation

To answer the first research question, linear regression models with high-level interactions were created in R iteratively, with different data included for each model. For the primary question, only the stream data from CastawaysWU was required. Building on the first research question, both the stream temperature and weather datasets were used to determine if these additional factors could improve a model's ability to predict stream temperature. Later, canopy coverage was also added to the model to determine if its factors could improve the model

performance. Multiple models were created for each new data source introduced, using different combinations of factors to analyze which columns have the most impact on stream temperature and to create a model that can explain the changes in the data.

Each model was evaluated using the standard graphs given by the `plot()` function in R along with the R-squared value, RMSE, and MAE. The residuals vs fitted graphs was used, as intended, to check the assumptions of linearity to ensure that a linear regression fits the data. Then, the normal Q-Q plot was used to check that the residuals created by the model were normally distributed ensuring that there is no systematic bias. The Scale-Location graph was used to check the homoscedasticity of the residuals. Finally, the Residuals vs Leverage to identify points that are impacting the model more than they should (outliers). In this stage of evaluation, three outliers were identified. Two of these points were recorded on the same day, and the third was the day prior. After additional research, these days were shown to be a part of a “freak, 10,000-year event,” according to author Monica Samayoa at OPB (2022). Thus, these points were removed from the dataset since they were deemed to not be representative of Oregon’s weather and stream temperatures. These plots ensured the models being created aligned with the required assumptions for linear regression and were appropriate for use.

While evaluating these models, primary focus was put on creating models with a high R-squared since the goal was a model that would explain the variation in the stream temperature reliably. However, the RMSE and MAE were also calculated based on the residuals for each model that was created. The goal was to create a model with an R-squared above 0.85, and an RMSE and MAE lower than 5 degrees fahrenheit. These metrics would signify that the chosen model can explain the variation in the datasets, and has low error in its predictions.

Results

- Actually what each model said, evaluation metrics, and how it relates to the research questions.

Discussion, Limitations, and Future Work

Discussion:

- What model we picked and why
- Implementations

Limitations:

The first challenge encountered through the course of this project was a lack of reliable data relevant to this research topic. Initially, a dataset on the Emerald Ash Borers had been identified that could be used to determine the beetle's impact on Salem's tree population. This dataset ended up being very challenging to web scrape to collect the data. The resulting data from the scrape only included 3 confirmed sightings of Emerald Ash Borers in Salem which would not have been enough data to add to the models that were created. There was also a dataset which contained every recorded tree in Salem, however, when this data was web scraped the results were unreliable. Thus, the project direction had to pivot multiple times as new data sources had to be found which had different contents. This left less time available for in-depth analysis and model creation which has left more work for the future.

- For this project and its use:
 - To predict the future stream temperature you must make assumptions about the weather.

- We are making big assumptions, interactions with precipitation and weather
- No time for arima model, would have used seasonality more reliably to predict into the future.

Future Work

- Include more reliable tree data
- Include EAB data when more is collected
- Include more streams in the analysis
 - And area upstream more
- Maybe use snowfall data in weather, the current one only had a measure of did it snow or not. Include ice events.
- Create the ARIMA model

Conclusions and Recommendations

- Tree canopy didn't prove to be significant, if more research was done and this was proven to be a consistent pattern then it suggests that the rising stream temperatures are due to another factor. Things like climate change should be looked into more.

Reproducibility

For full access to the repository where the project's products—data, markdowns, and other supporting documents—are stored, visit [the Project Github](#). Additional information about the process used and project milestones can be found on the [Github Project Board](#).

To reproduce this project, Python 3.11.1, R 4.5.2, Beekeeper SQL 5.9, and Railway were used for data modification and storage. Within R and Python, the libraries that were used is listed

at the top of the notebooks and markdowns, but is also included here: NumPy, Matplotlib, Seaborn, Pandas, scikit-learn, tidyverse, ggplot2.

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